



Netramind

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1 PURPOSE

This SOP outlines the administration and maintenance of computerized systems used by Netramind to support GxP activities, ensuring systems remain controlled, secure, and fit for intended use.

2 SCOPE

This procedure applies to GxP-relevant computerized systems maintained or used by Netramind, including internally managed and externally hosted systems.

3 REFERENCES

- [SOP-010 Software Development Lifecycle](#)
- [SOP-008 Incident Management](#)
- [SOP-014 Computer System Validation](#)
- [SOP-017 Audit Trail Review](#)
- [SOP-025 Periodic Review of IT Systems](#)
- [SOP-019 User Access Management](#)
- 21 CFR (Code of Federal Regulations) PART 11 Electronic Records; Electronic Signatures

4 DEFINITIONS

- 4.1 Audit Trail—Metadata records that allow the appropriate evaluation of the course of events by capturing details on actions (manual or automated) performed relating to information and data collection and, where applicable, to activities in computerized systems. The audit trail should show activities, initial entry and changes to data fields or records, by whom, when and, where applicable, why. In computerized systems, the audit trail should be secure, computer-generated and time stamped.
- 4.2 Corrective and Preventive Action (CAPA)— A systematic approach that includes actions needed to correct (“correction”), prevent recurrence (“corrective action”), and eliminate the cause of potential nonconforming product and other quality problems (preventive action).
- 4.3 Change Control—A general term describing the process of managing how changes are introduced into a controlled system.
- 4.4 Computerized System—The set of hardware, software, peripheral devices, personnel, and documentation needed to perform a specific group of functions to meet a business or regulatory need. A Computer System (defined above) is a component of a Computerized System.
- 4.5 Data Acquisition Tool (DAT) — A paper or electronic tool designed to collect data and associated metadata from a data originator in a clinical trial according to the protocol and to report the data to the sponsor. The data originator may be a human (e.g., the participant or trial staff), a machine (e.g., wearables and sensors) or a computer system from which the electronic transfer



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of data from one system to another has been undertaken (e.g., extraction of data from an electronic health record or laboratory system). Examples of DATs include but are not limited to CRFs, interactive response technologies (IRTs), clinical outcome assessments (COAs), including patient-reported outcomes (PROs) and wearable devices, irrespective of the media used.

- 4.6 End User— The final or ultimate user of a computer system. An end user is an individual who uses the product for its intended purpose after it has been fully developed and marketed. A System Administrator is not considered to be an End User.
- 4.7 Firmware— A type of software program embedded into hardware devices to help the devices' function, so the combination of a hardware device and computer instructions/data that resides as read only software on that device. The software may or may not be configurable. The computer cannot modify such software during processing.
- 4.8 Incident Report (IR). — An official report of an identified incident found in the computerized system.
- 4.9 Interface— A system, device or other entity that provides input or output for a computerized system but is not considered to be within the design boundary of the computerized system.
- 4.10 Metadata— The contextual information required to understand a given data element. Metadata is structured information that describes, explains or otherwise makes it easier to retrieve, use or manage data. Relevant metadata are those needed to allow the appropriate evaluation of the trial conduct.
- 4.11 Production Environment—This is the environment used for normal business use. The system is used to create, store, modify and or transfer GxP data. The environment is controlled so that all changes to the environment are documented.
- 4.12 Regression Testing—A test performed meant to verify that software previously developed and working is still working correctly after it was changed.
- 4.13 Release Management —The process of managing, planning, scheduling and controlling software builds through different stages and environments.
- 4.14 Root Cause Analysis (RCA)— The process of identifying the causal factors of noncompliance, a deviation, violation, or undesirable outcome.
- 4.15 Service Level Agreement (SLA)— A contract between a service provider and customer that defines the service to be provided, and the level of performance expected and measured. SLAs are typically in place for IT service providers.
- 4.16 Software as a Service (SaaS) – Applications which are hosted by a service provider and are made available to users on a subscription basis, eliminating the need for local installation and infrastructure management. Netramind still is responsible for qualifying the service provider of SaaS applications including verification that IQ and OQ were sufficiently executed by the SaaS service provider and performing a risk-based validation based on user requirements, as the



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regulatory responsibility for ensuring and demonstrating the integrity of GxP data in the SaaS solution remains with Netramind.

- 4.17 Software Development Lifecycle (SDLC) — The model that defines the phases and procedures for the process of defining, developing, testing, maintaining, and retiring systems. Several phases exist in the SDLC, which support successful software development and maintenance. The phases are definition, development, validation, and production/maintenance.
- 4.18 System Administrator— Individual who has the functional responsibility for the system; for providing the data or function
- 4.19 System Owner— Individual who has the functional responsibility for the system; for providing the data or function that the computerized system addresses, known as user requirements. It is possible that different parts of a complex integrated system would have multiple System Owners.
- 4.20 Traceability Matrix (TM) — A matrix that maps the specification requirements to the test scripts verifying that the specification was met.
- 4.21 Uninterruptable Power Supply (UPS) — A device that allows a computer to keep running for at least a short time when the primary power source is lost. It also provides protection from power surges.
- 4.22 Validation— Establishing documented evidence that provides a high degree of assurance that a specific computerized system will consistently operate in accordance with pre-determined specifications and continues to operate as specified throughout its system lifecycle.
- 4.23 Validation Plan (VP)— A formal document providing a framework that describes the validation process, responsibilities, documentation deliverables, timelines, acceptance criteria and maintenance of the validated state for a given computerized system. The VP encompasses the system's lifecycle (development, implementation, use and retirement).
- 4.24 Validation Summary Report (VSR) — A document that summarizes the validation activities, exceptional conditions occurring during testing and the justification for releasing for use or continued use of the system.

5 RESPONSIBILITIES

5.1 System Owner

- 5.1.1 Maintain the validated state of the system.
- 5.1.2 Review and approving validation documents as required.
- 5.1.3 Ensure changes to a validated system are performed via formal change control procedures.
- 5.1.4 Ensure specification documentation is up to date, and consistent with the needs of the system's users.
- 5.1.5 Define scope of maintenance releases.



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- 5.1.6 Develop and/or update SOPs for the intended use of the validated system.
- 5.1.7 Ensure the performance of validation activities according to this SOP, including the conduction and documentation of validation testing.

5.2 Users

- 5.2.1 Follow change control procedures.
- 5.2.2 Report system problems or failures to the System Owner or IT as appropriate.
- 5.2.3 Support validation activities as needed.

5.3 Quality Assurance (QA) Validation

- 5.3.1 Review and approve validation activities and ensure compliance.
- 5.3.2 Manage incident tracking.
- 5.3.3 Review and approve all testing and validation documents.
- 5.3.4 Summarize the results of all testing in a VSR.
- 5.3.5 Support Production Deployment.
- 5.3.6 Evaluate systems for compliance with applicable and current regulations for computerized systems and data integrity.

5.4 Development Team

- 5.4.1 Prepare and/or maintain up to date functional and design requirements to satisfy the user's requirements.
- 5.4.2 Develop the system to satisfy the required specifications.
- 5.4.3 Execute peer code review, unit and integration tests following development best practices and standards.
- 5.4.4 Comply with source and release management policies and procedures.
- 5.4.5 Support testing, validation, production deployments, and post-production deployment.

5.5 Technical Team/Information Technology Team (IT)

- 5.5.1 Provide technical infrastructure as per system requirements.
- 5.5.2 Responsible for installing and configuring system environments.
- 5.5.3 Perform release management activities.
- 5.5.4 Provide technical support during validation activities.
- 5.5.5 Maintain the system in accordance with its validated state.
- 5.5.6 Review and approving validation documents as required.
- 5.5.7 Evaluate any system related risks and errors during implementation of the system and post-production deployment.

6 PROCEDURE

6.1 General System Administration and Support



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- 6.1.1 All support and change requests for IT systems, Netramind-developed or SaaS based services are managed through a Change Request process according to **FRM-021-01 IT Change Control**.
- 6.1.2 Account Management and Access Control
 - 6.1.2.1 User access, change/update and removal of access procedures are described in **SOP-019 User Account Management**.
- 6.1.3 System Release Updates
 - 6.1.3.1 System updates from internal or external providers must be assessed and managed through change control, as appropriate.
 - 6.1.4 IT performs any standard change to the specific system following **FRM-021-01 IT Change Control**. Standard changes include changes to controlled terminology and/or routine maintenance.
 - 6.1.5 Major changes to the specific system adhere to the procedures defined in SOP-ABC-123 Software Development Lifecycle.
- 6.1.6 IT Help Desk Support
 - 6.1.6.1 Netramind utilizes the IT Help Desk function, either resourced internally or delegated to a third-party technology support service provider, to manage requests related to technology, systems software and hardware components.
 - 6.1.6.2 The IT Help Desk technical staff is responsible for evaluating, resolving, tracking and closing help desk support tickets.
- 6.1.7 IT Service Level Agreement (SLA)
 - 6.1.7.1 SLAs are established between service/technical providers for systems which are not developed or managed internally by Netramind IT. Security expectations such as authentication controls (e.g., MFA) should be defined where applicable.
 - 6.1.7.2 IT and System Owner are responsible for the relationship with the providers, including reviewing and approving the criteria defined in the SLA, including expectation for system administration and system maintenance.
 - 6.1.7.3 IT and System Owner are responsible for the oversight of the service provider to ensure tasks are executed according to the criteria agreed to in the SLA.
- 6.2 System Maintenance Planning
 - 6.2.1 The System Owner reviews all approved change requests to current operational system and open incidents to define maintenance releases.
 - 6.2.2 All maintenance releases and deliverables must comply with procedures defined in **SOP-010 Software Development Lifecycle**.



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6.2.3 Change Impact Assessment

- 6.2.3.1 Changes must be assessed for impact on system functionality, data integrity, and validation status. Appropriate testing and documentation updates must be defined based on this assessment.

6.2.4 Technical Change Development

- 6.2.4.1 The changes are built per the specifications, Change Control Request (CCR) and/or Incident Report (IR).
- 6.2.4.2 All technical changes to the system are documented in development notes for each **FRM-021-01 IT Change Control Form** and/or **FRM-021-02 Incident Request Form**. All changes must follow applicable SOPs on SDLC and Incident Management.

6.2.5 Change Control

- 6.2.5.1 Change Control ensures that proposed changes to computer systems will be properly documented, reviewed, evaluated, approved or disapproved/rejected, and implemented, and that approved changes shall be communicated to the appropriate personnel in a timely manner.
- 6.2.5.2 Execute change control in accordance with **FRM-021-01 IT Change Control Form**.

6.2.6 Validation

- 6.2.6.1 System testing will be performed to ensure there are no major issues with the system prior to releasing the system for validation & qualification testing.
- 6.2.6.2 Validation & qualification testing will be executed following the standards in **SOP-014 Computer System Validation**.

6.2.7 Regression Testing

- 6.2.7.1 Regression testing is performed to satisfy the tasks identified in the Validation Plan (VP).
 - Rerun identified test cases previously executed
 - Identify any unintended effects of the system changes

6.3 Technical Release Management

- 6.3.1 The system release plan is determined in accordance with procedures described in **SOP-010 Software Development Lifecycle**.

6.4 Incident Management

- 6.4.1 All identified incidents during technical change implementation are properly documented, reviewed, approved or rejected, and fixes are planned according to severity



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and priority. Incident Management is performed in accordance with **SOP-008 Incident Management**.

6.5 Hardware Maintenance

6.5.1 Hardware supporting GxP systems must be maintained through preventative maintenance, monitoring, and timely updates to ensure reliability and performance.

6.6 Periodic Reviews

6.6.1 Review of specific system requirements, processes are specified in the Validation Plan and should be incorporated into the planned periodic reviews of the systems, for example: issues are discovered during the Periodic review, a gap analysis will be performed to establish a remediation plan. The remediation plan will be executed according to **FRM-021-01 IT Change Control Form** procedure.

6.6.2 During the Maintenance Phase all changes to the system follow **FRM-021-01 IT Change Control Form I** and **SOP-008 Incident Management** procedures and must be approved by the System Owner and QA.

6.6.3 On a pre-determined frequency, IT and System Owner perform periodic reviews of system requirements, validation, user access and audit trails/metadata.

6.6.3.1 Periodic reviews of systems are conducted as described in **SOP-025 Periodic Review of IT Systems** and specific system's validation plan (VP), as appropriate. Reviews will include the following:

- Confirmation that the version of the software in use is the same as documented
- Validation documentation is adequate to represent the actual system
- Review of change controls since the last periodic review
- System is being used per SOPs
- Review of system security including assessing that only the appropriate users have access to the system

6.6.3.2 Periodic reviews of user access management, identification coder and passwords are managed in accordance with **SOP-019 User Access Management**.

6.6.3.3 Periodic reviews of system's metadata including audit trails follow procedures described in **SOP-017 Audit Trail Review**.

6.6.4 Deviations, deficiencies identified during periodic reviews must be documented, impact assessment is conducted and as appropriate, managed through the procedures for



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incident reporting per **SOP-008 Incident Management**, and/or corrective actions **SOP-007 Corrective and Preventive Action (CAPA)**.

6.7 System Backup/Restore

6.7.1 GxP system backup and restore processes managed by external technical service providers are maintained in accordance with service level agreement (SLA) between Netramind and the service provider, and service provider's procedures.

6.8 Disaster Recovery (DR)

6.8.1 GxP system disaster recovery processes managed by external technical service providers are maintained in accordance with service level agreement (SLA) between Netramind and the service provider, and service provider's procedures.

6.8.2 DR testing and assessment of Netramind GxP systems are handled in accordance with **SOP-023 IT Disaster Recovery**.

6.9 Business Continuity (BC)

6.9.1 BC testing and assessment of Netramind GxP systems are handled in accordance with **SOP-025 Business Continuity**.

6.9.2 In case of unavailability of the system(s) for extended period, Netramind may delegate or outsource GxP activities to service providers as necessary.

6.10 System Decommissioning

6.10.1 When it is determined that a system is to be obsoleted, replaced or upgraded, the action of retiring the system must be initiated through change control and adhere to procedures described in **SOP-010 Software Development Lifecycle**.

6.10.2 All validation documentation will be filed and retained until all data associated with the system has been destroyed in accordance with regulations.

6.11 Documentation

6.11.1 Review documentation to determine which documents have been impacted by a change. All approved documents that have been affected should be updated in accordance with the SOP Configuration Management.

6.11.2 Verify all supporting documents have been updated to reflect the changes.

6.11.3 Maintain all supporting documents and have accessible for reference during maintenance.

7 ATTACHMENTS

- **FRM-021-01 IT Change Control Form**



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- FRM-021-02 Incident Request Form.

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8 REVISION HISTORY

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SIGNATURE PAGE

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Author:

NAME	TITLE	SIGNATURE	DATE
Justin Shaka	Fractional COO	Digitally signed by Justin Shaka Name : Justin Shaka Email : justin@netramind.ai Reason : I reviewed and agree to this document Time : May 01 2026 13:28 EDT Signed with : Zoho Sign C798418707456778BEE159BD1D9FFED08...	01-MAY-2026

Reviewer:

NAME	TITLE	SIGNATURE	DATE
Jay Chhablani	President	Digitally signed by Jay Chhablani Name : Jay Chhablani Email : chhablani@netramind.ai Reason : I approve this document Time : May 01 2026 14:11 EDT Signed with : Zoho Sign 6D081372BA602642A6FC5157FF390243ED...	May 01 2026

Approver:

NAME	TITLE	SIGNATURE	DATE
Sandeep Chandra Bollepalli	Director of AI	Digitally signed by Sandeep Chandra Bollepalli Name : Sandeep Chandra Bollepalli Email : sandeep@netramind.ai Reason : I approve this document Time : May 01 2026 14:19 EDT Signed with : Zoho Sign 6877F16A7089178BFD3212614822B8995AE...	May 01 2026